

Antifouling behaviour of pH responsive amphiphilic smart coatings on the basis of charge conversion in the marine environments

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ABSTRACT

In response to complex marine fouling issues, this study developed pH-responsive amphiphilic antifouling coatings (PDHM) synthesized via condensation reactions between protonated zwitterionic-tertiary amine polymers and polydimethylsiloxane (PDMS). These amphiphilic coatings demonstrate pH-responsive surface charge transitions governed by protonation equilibria. Under seawater conditions (pH $\sim$ 8.3), the PDHM surface exhibits negative charge enrichment, synergizing with a hydrated interfacial layer to impede biofouling adhesion. Subsequent surface acidification induced by colonizing organisms triggers protonation of tertiary amines, converting the surface charge to positive. This transition disrupts fouling organism viability and promotes release under hydrodynamic shear forces. Relative to PDMS, PDHM coatings achieved 99.54 % and 99.59 % reduction in bacterial adhesion (*B. subtilis* and *P. rutenica*, respectively), alongside a 98.23 % reduction in algal adhesion (*P. rutenica*). Marine field evaluations in the East China Sea confirmed sustained antifouling efficacy for 180 days. Furthermore, 16S rDNA analysis revealed significantly attenuated biofilm formation and reduced abundance of mobile genetic elements (MGEs) on PDHM surfaces versus PDMS, corroborating its superior antifouling performance. This work establishes a novel methodology for designing smart-responsive marine antifouling coatings.

1. Introduction

Marine biofouling constitutes a pervasive phenomenon of biological adhesion [1,2], characterized by the accumulation of bacteria, algae, and crustaceans on submerged maritime structures [3,4], including vessels and aquaculture infrastructure. This colonization increases structural mass, impairs hydrodynamic performance, accelerates bio-corrosion, and elevates safety risks, thereby significantly impeding marine economic development [5–8]. Consequently, researchers have developed antifouling coatings to mitigate organismal adhesion, exemplified by fouling-release coatings (FRCs) that minimize interfacial adhesion strength [9–13]. Nevertheless, the operational efficacy of FRCs remains constrained by dependence on hydrodynamic conditions and nonspecific adsorption phenomena inherent to hydrophobic surfaces [13,14]. To address these limitations, hydrophilic constituents such as poly(ethylene glycol) (PEG) [15,16] and zwitterionic polymers [17,18]

have been incorporated into PDMS-based systems. Through chemical cross-linking strategies, these modifications yield amphiphilic coatings exhibiting nanoscale chemical heterogeneity. This engineered heterogeneity disrupts fouling mechanisms by confounding organismal settlement and imposing entropic penalties during adhesion processes, thereby substantially enhancing antifouling performance [19].

PEG and its derivatives represent widely studied hydrophilic materials renowned for their effective protein resistance. These polymers establish robust hydration layers through hydrogen bonding with water molecules, elevating the energy barrier for fouling organism adhesion and consequently inhibiting biofouling [20]. Despite these merits, PEG-based systems exhibit limitations including susceptibility to oxidative degradation and functional group deactivation, compromising long-term stability [21]. To advance the development of durable amphiphilic coatings, zwitterionic polymers have emerged as promising alternatives to PEG. Zwitterions-electroneutral macromolecules

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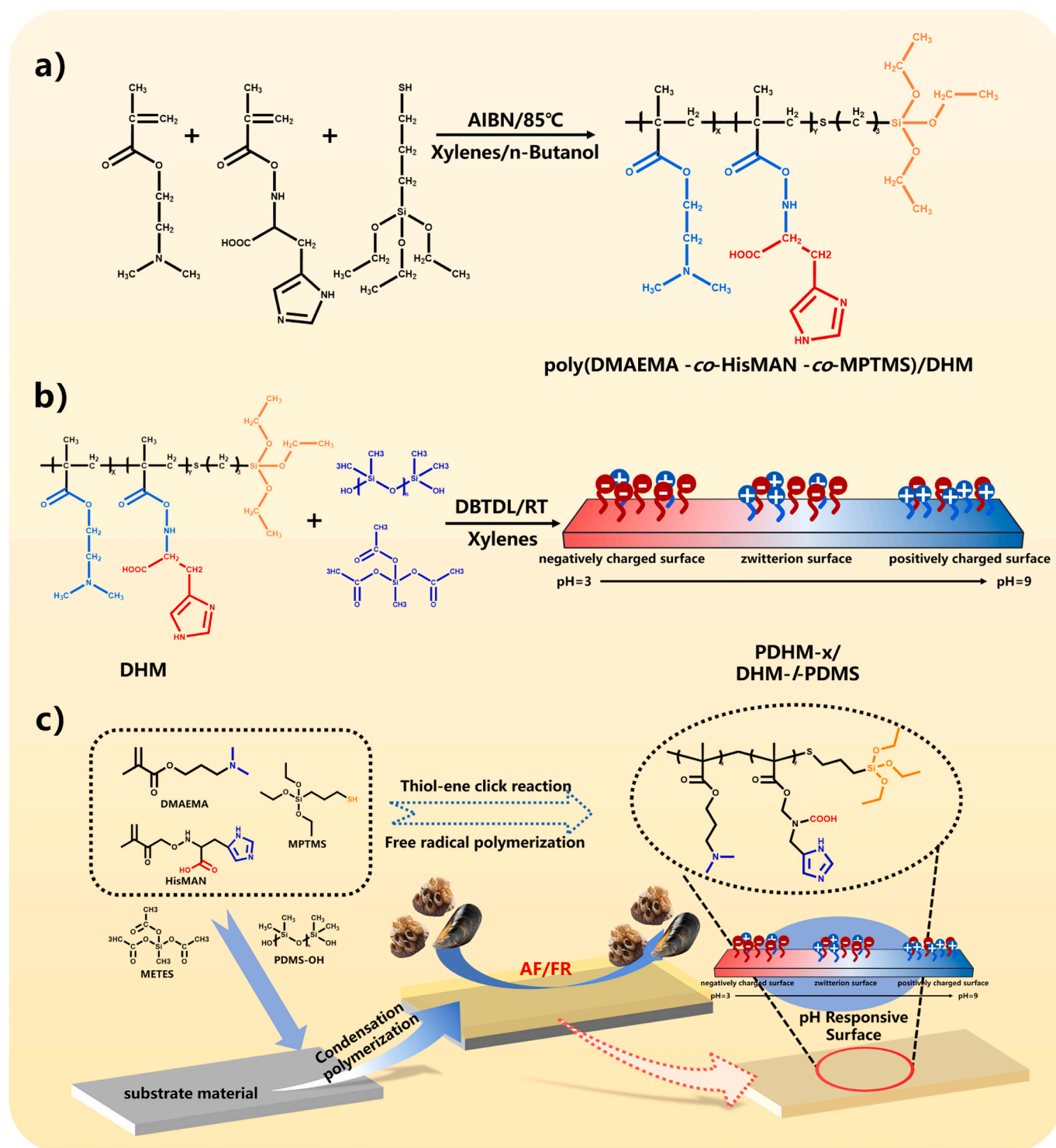
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incorporating stoichiometric, uniformly distributed cationic and anionic moieties-generate ultra-dense hydration layers via ionic solvation. Relative to the hydrogen-bonded hydration of nonionic polymers (e.g., PEG), zwitterion-induced hydration demonstrates substantially greater hydration density [22], endowing surfaces with exceptional protein repellency, interfacial lubrication, and antifreeze characteristics [23]. Illustrating this approach, Dai et al. [24] synthesized a zwitterionic copolymer through free-radical ring-opening copolymerization of methyl methacrylate, a zwitterionic tert-carboxybetaine triisoacrylate

monomer, and 2-methylene-1,3-dioxolane. The strategic incorporation of hydrolyzable ester bonds within the polymer backbone facilitated segmental degradation and autonomous surface renewal under static immersion. This self-renewing coating exhibited significant resistance against adhesion by marine bacteria (*Pseudomonas* sp.) and diatoms (*Navicula incerta*), demonstrating potential for sustained antifouling efficacy.

Zwitterionic polymers impart effective antifouling and antimicrobial functionality through synergistic hydration layers and dynamic charge



Scheme 1. (a) Preparation of pH-responsive hydrophilic polymer chain segments containing zwitterion-DMAEMA, (b) Preparation of amphiphilic smart antifouling coating PDHM-x, (c) Schematic of the anti-fouling mechanism and pH response of the PDHM coating.

modulation, while exhibiting pronounced anti-polyelectrolyte effects and ionic specificity. These intrinsic properties enable environmentally responsive behaviors, forming the basis for intelligent antifouling systems [25–27]. Illustrating this, Zhang et al. [28] engineered biocompatible zwitterionic surfaces with pH-responsive antimicrobial functionality. These surfaces transition from fouling-resistant zwitterionic states to bactericidal cationic states upon environmental acidification, a process triggered by bacterial colonization. Initial surface hydration prevents microbial adhesion, subsequent microbial metabolism acidifies the local microenvironment, cleaving amide bonds to convert zwitterions into quaternary ammonium compounds. Confocal laser scanning microscopy and SEM analyses confirmed this transformation within acidic microdomains, conferring dual anti-adhesive and bactericidal capabilities. This charge conversion mechanism operates via dynamic bond cleavage, where anion elimination generates net-positive surfaces for contact-killing. Building on this concept, Sun et al. [29] integrated pH-responsive poly(carboxybetaine methacrylate-co-2-(dimethylamino)ethyl methacrylate) (P(CBMA-co-DMAEMA)) with antimicrobial peptides, creating a sequentially activated antimicrobial system. The zwitterionic surface initially resists bioadhesion, following colonization-induced acidification, the resultant positive zeta potential facilitates electrostatic microbial adsorption while concurrently releasing tethered antimicrobial peptides for targeted eradication. Beyond established efficacy against proteins, bacteria, diatoms, and spores [30,31], zwitterionic systems can be further engineered for environmental intelligence via pH-dependent protonation of carboxyl/amine functionalities. Despite extensive mechanistic studies on pH responsiveness, its application in marine antifouling contexts remains comparatively underexplored.

Marine ecosystems harbor substantial dissolved organic carbon (DOC) reservoirs [32], serving as the primary carbon substrate for fouling communities (e.g., heterotrophic microorganisms). During DOC catabolism, these organisms excrete acidic metabolites—including polysaccharides, organic acids, and toxins—that locally acidify their microenvironment. Capitalizing on this ecological trigger, we engineered a chemically heterogeneous, pH-responsive antifouling coating by copolymerizing histidine methacrylamide (HisMAN) and 2-(dimethylamino)ethyl methacrylate (DMAEMA) via chain-transfer polymerization with 3-mercaptopropyltriethoxysilane (MPTMS), yielding the hydrophilic DHM polymer (Scheme 1a). Critically, stoichiometric modulation of DMAEMA/HisMAN leverages their distinct pKa values to enable dynamic surface charge reversal—a design innovation exploiting ecological feedback for on-demand antifouling. Subsequent silanol condensation with hydroxyl-terminated polydimethylsiloxane (PDMS-OH) generated the PDHM coating (Scheme 1b).

In marine environments (pH ~ 8.3), the surface exhibits:

- (i) Net-negative charge and a robust hydration layer, preventing initial biofouling adhesion;
 - (ii) Triggered transformation upon acidic metabolite secretion:
- a) Charge reversal to net-positive, disrupting microbial electrochemical homeostasis,
 - b) Exposure of hydrophobic domains, enabling foulant release under hydrodynamic shear.

To validate the mechanism, we pioneered comparative 16S rDNA sequencing of incipient colonization on PDHM versus PDMS surfaces. This work establishes an integrated framework combining stimuli-responsive materials with genomic verification for intelligent marine antifouling control, offering an ecologically informed paradigm shift in mitigation technology.

2. Materials and method

2.1. Materials

Dihydroxyl-terminated polydimethylsiloxane (HO-PDMS-OH, Mw = 28,000 g/mol) was purchased from Wacker. Xylene, L-histidine, methacryloyl chloride, K₂CO₃, 3-Mercaptopropyltriethoxysilane (MPTMS), 2-(Dimethylamino) ethyl methacrylate (DMAEMA) and Methyltriacetoxysilane (METES) were purchased from Macklin. NaNO₂, n-butanol, n-hexane, Dibutyltin dilaurate (DBTDL) and 2,2'-Azobisisobutyronitrile (AIBN) purchased from Aladdin. The water used in all of the experiments was purified using the MOLUV30D system. All reagents were used as received. All of the chemicals used in this experiment were of analytical grade, with a purity of at least 99.7 %.

2.2. Synthesis of poly(DMAEMA-co-HisMAN-co-MPTMS)(DHM)

The random copolymer poly(DMAEMA-co-HisMAN-co-MPTMS) (DHM) was synthesized via concurrent thiol-ene click chemistry and free radical polymerization using azobisisobutyronitrile (AIBN) as initiator (Scheme 1a). DMAEMA (7.86 g, 50 mmol), AIBN (0.05 g, 0.3 mmol), and HisMAN (mass specified in Table S1; preparation detailed in Supporting Information [33]) were dissolved in a xylene/butanol mixture (15 mL, 2:1 v/v) under ultrasonic agitation to form homogeneous Solution A. This solution was added dropwise (0.5 mL/min) to a nitrogen-purged reactor containing anhydrous xylene (5 mL). Following degassing via N₂ sparging (15 min), the reaction mixture was heated to 85 °C and maintained under reflux for 12 h. Subsequently, MPTMS (1.50 g, 6 mmol) and additional AIBN (0.05 g, 0.3 mmol) dissolved in anhydrous xylene (5 mL) were introduced dropwise. Polymerization continued under anaerobic conditions at 85 °C for 4 h. The reaction mixture was cooled to ambient temperature, and solvents were removed by rotary evaporation to yield a viscous yellow-brown crude product. Unreacted monomers and initiator residues were extracted through triple precipitation in hexane (200 mL). The purified polymer was desiccated under vacuum (30 °C, 24 h) to constant mass. Gel permeation chromatography (GPC) analysis determined the molecular weight distribution, Mn = 14,770 g/mol, Mw = 26,492 g/mol, with polydispersity index (PDI) = 1.79.

2.3. Coating preparation

The amphiphilic PDHM coating was fabricated via silanol condensation crosslinking as illustrated in Scheme 1b. Precursor components DHM and hydroxyl-terminated polydimethylsiloxane (PDMS-OH) were formulated at prescribed mass ratios (Table S2) and dispersed in xylene/butanol (10 mL, 7:3 v/v) under ultrasonic treatment. This yielded homogeneous Solution B, which underwent continuous mechanical agitation (30 min, 25 °C). Subsequently, the crosslinking agent 3-(methylamino)propyltriethoxysilane (METES) and catalyst dibutyltin dilaurate (DBTDL) were dissolved in anhydrous xylene (5 mL), ultrasonicated (5 min), and introduced dropwise (0.5 mL/min) into Solution B. The reaction mixture was agitated (30 min) to afford a colloidal suspension designated DHM-*l*-PDMS, hereafter termed PDHM-*x* (where *x* denotes DHM mass fraction: 5, 10, or 15 wt%). The resultant product was deposited onto substrates via controlled coating (200 μm thickness) and cured (24 h, 25 °C) to generate silicone-based amphiphilic coatings exhibiting pH-responsive functionality through incorporated DHM domains.

2.4. Characterization

Fourier transform infrared spectroscopy (FT-IR) and attenuated total reflectance infrared spectroscopy (ATR-IR) data were analyzed using a Nicolet iS50 spectrometer (Thermo, Waltham, MA, USA) in the 4000–400 cm⁻¹ scanning range. The zeta potential of the polymers was

tested at different pH values (pH = 3.0, 4.0, 5.0, 6.0, 7.0, 8.0 and 9.0) using a nanoparticle size potential analyser (NanoBrook 90 Plus PALS, Brookhaven, NY, USA) to investigate the response intervals of the pH-responsive polymers. To ensure the reliability of the test, the ambient temperature and pH were kept stable during the test and three replicates of the test were performed for each sample to obtain the zeta potential of the coated surfaces at different pH values. The 3D morphology of the surface of the sample was analyzed by means of a confocal laser scanning microscope (CLSM, KEYENCE VK-X250, KEYENCE, Osaka, Japan). To measure the surface morphology of the coating, the XYZ fast scanning mode was used. The adhesion of the coatings to the substrate surface was determined by the pull-off test according to ASTM D4541–17. PDHM-x and PDMS were coated on the surface of epoxy resin substrate boards coated with intermediate paint, respectively, and the dry film thickness of the coated coatings was $250 \pm 30 \mu\text{m}$. After curing for 24 h at room temperature, the coatings were measured using a peel adhesion tester (Posi Test AT-A, DeFelsko, NY, USA). To eliminate errors, the test was repeated three times on different parts of the same coating and the average value was taken. The impact resistance of the samples was assessed using a paint film impact tester (TCJ-II, Shanghai Modern Environmental Engineering Co., Shanghai, China) in accordance with ISO-6272-1 (2011). Paint adhesion was evaluated according to ASTM D3359–17. The contact angle measuring instrument (Theta Flex, Biolin, Sweden) was used for the quantification of the contact angle between water and diiodomethane in air. Water contact angle measurements used 3 μL distilled water, while diiodomethane contact angle measurements used 2 μL diiodomethane. Five different positions on each sample were evaluated and the results were averaged to obtain the mean. The Owens-Wendt method (Table S3) was used to determine the surface free energy of the coating. After immersion in artificial seawater for 3 days, the water dynamic contact angle was measured. A drop of water (3 μL) was applied to the sample surface at room temperature. Subsequently, the water contact angle was meticulously recorded at 0, 1, 3, 5, 7 and 10 min, respectively.

2.5. Bacterial biofouling resistance test

The anti-bacterial adhesion performance of the coating was evaluated by selecting *Bacillus subtilis* (*B. subtilis*) and *Pseudoalteromonas ruthenica* (*P. ruthenica*) as representative experimental bacteria. The coating was then placed into a 24-well cell culture plate, after which 100 μL of bacterial suspension (10^6 CFU/mL) was added. Subsequent to an incubation period of 48 h at 37°C , the surface was gently washed on three occasions with ASW in order to remove any bacteria which had not become firmly attached. Subsequently, the inverted fluorescence microscope was employed for the observation and capture of images. Finally, the area of bacterial attachment was quantified using ImageJ. Furthermore, PBS solutions with pH values of 4.0 and 7.0 were added under the aforementioned conditions, respectively, and then compared with the ASW environment (pH = 8.3). This was done to investigate the resistance to bacterial adhesion on the surface of the PDHM coating under different pH environments.

Initially, the adhesion areas of *B. subtilis* and *P. ruthenica* on the PDMS surface were quantified as N_1 . The adhesion area of these bacteria on the PDHM-x surface was then measured as N_2 . The samples employed in the aforementioned experiment were subjected to five separate tests, with the mean value documented as the experiment's result. The anti-adhesion rate (R) was subsequently determined using Eq. (1):

$$R = \left(1 - \frac{N_2}{N_1}\right) \times 100\% \quad (1)$$

As illustrated in Table S5, the extent of coverage and anti-adhesion rate of bacteria on the coating is presented [34].

2.6. Algae anti-adhesion test

Phaeodactylum tricornutum (*P. tricornutum*) was selected for the algae anti-adhesion test. Initially, the samples were immersed in a marine microalgae culture medium, followed by the addition of 30 μL ($5 \cdot 10^6$ cell/mL) of algae suspension for cultivation under light-dark cycle conditions at 20°C for a duration of 7 days. Thereafter, the surface was subjected to a triple rinse with sterile ASW, in order to ensure complete elimination of all extant algal cells. Finally, adhesion was observed using a fluorescence inverted microscope and the adhesion area was quantified using ImageJ.

P. tricornutum counts following water treatment of the PDMS coating were documented as N_3 , whilst those of the PDHM-x coating were recorded as N_4 . Five fluorescence images of each sample were obtained to ascertain the anti-adhesion rate (R), which was subsequently calculated using Eq. (2):

$$R = \left(1 - \frac{N_4}{N_3}\right) \times 100\% \quad (2)$$

As outlined in Table S5, the extent of coverage and anti-adhesion rate of algae on the coating have been demonstrated [34].

2.7. Marine field test

The antifouling performance of the amphiphilic coating (PDHM-x) is examined through the application of the sample onto epoxy resin plates ($100 \times 80 \times 3 \text{ mm}$). The thickness of the coating on the sample was measured using a film thickness gauge, revealing a thickness of $220 \pm 30 \mu\text{m}$. The coated samples underwent immersion in the Xiamen-sea area of the East China Sea ($24^\circ 33'\text{N}$, $118^\circ 09'\text{E}$) at a depth of 0.5–2.0 m over a duration of 180 days during the sea trial (18th October 2023 to 18th April 2024). During the experiment, the temperature of the seawater ranged from $29.08 \pm 0.59^\circ\text{C}$ to $16.94 \pm 0.42^\circ\text{C}$, while the pH value ranged from 8.20 ± 0.035 to 8.03 ± 0.063 . At regular intervals, the coating was retrieved from the sea, thoroughly rinsed with seawater, and photographed for further analysis. The experiment will be conducted in accordance with the protocol outlined in GB/T 5370–2007 to assess the antifouling property and calculate the resultant score.

3. Results and discussion

3.1. Characteristics of the coating

3.1.1. Chemical structure and wettability

The molecular structure of HisMAN has been characterized and analyzed by FT-IR as shown in Fig. S1. The FT-IR spectra of HisMAN were found to be in agreement with those previously reported in the scientific literature [35], in which the single peak of N–H stretch vibration at 3300 cm^{-1} , the absorption peak attributed to alkene-hydrogen at 3080 cm^{-1} , the C–H absorption peak at 2975 cm^{-1} , and the scissoring vibration of the C–H at 1466 cm^{-1} were observed, respectively. Furthermore, the stretch vibration peak of the amide II absorption bands (C=O) can be observed at 1680 cm^{-1} , which is hypsochromic shift due to the n- π conjugation of the lone pair of electrons on the nitrogen atoms in the secondary amine group with the carbonyl group. As the olefinic double bond absorption peak is weak and occurs at a close proximity to the carbonyl group, it is not observable in FT-IR. It is evident that the aforementioned analytical process is conducive to the effective synthesis of HisMAN. Subsequently, HisMAN, DMAEMA and MPTMS were utilized in the synthesis of DHM through the concurrent application of thiol-ene click chemistry and free radical polymerization. The characterization of the compound was conducted via FT-IR, with the resulting spectrum depicted in Fig. 1a. The FT-IR of DHM and DMAEMA-co-MPTMS was compared, revealing that stretch vibrations peaks attributed to C=O, C–N, Si–O–C and C–S at 1730 cm^{-1} , 1280 cm^{-1} ,

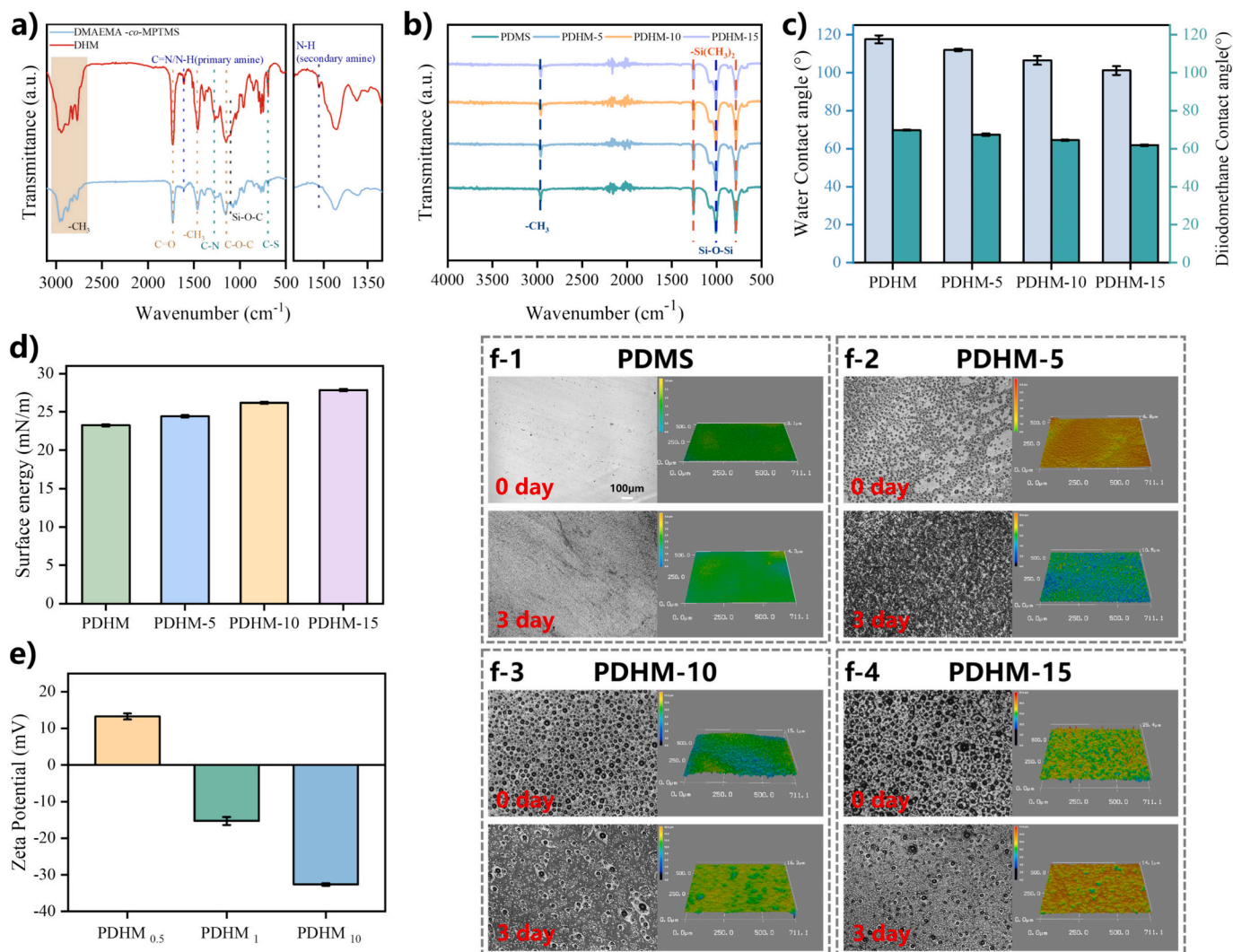


Fig. 1. (a) Comparison of FT-IR spectra of DHM and DMAEMA-co-MPTMS, (b) comparison of ATR-IR spectra of PDMS and PDHM-x, (c) water contact angle and diiodomethane contact angle of PDHM-x coatings, (d) surface Energy of PDHM-x Coating, (e) the zeta potential of PDHM coatings prepared with different HisMAN/DMAEMA ratios under neutral conditions, and (f1–4) Visual and 3D images of PDHM-x coating surface morphology before and after immersion.

1098 cm^{-1} and 689 cm^{-1} , respectively, for both polymers. In the DHM sample, peaks corresponding to the N–H deformation vibrational spectrum of primary amine at 1600 cm^{-1} or the C=N stretch vibrational spectrum of imidazole, as well as the N–H deformation vibrational spectrum of secondary amine at 1516 cm^{-1} , are discernible. This finding indicates that the absorption peaks attributed to imidazole and amide groups in HisMAN can be observed in DHM, thereby providing further evidence that the preparation of DHM has been successfully achieved. Fig. 1b shows the ATR-IR patterns of the PDMS and PDHM-x coatings. By comparison, it can be seen that the stretch vibrations peaks attributed to $-\text{Si}(\text{CH}_3)_2$, $\text{Si}-\text{O}-\text{Si}$ and $-\text{CH}_3$ were observed at 861 cm^{-1} , 1260 cm^{-1} , 1003 cm^{-1} and 2960 cm^{-1} for both the PDMS and PDHM-x coatings. In addition, no other specific absorption peaks were observed for the PDHM-x coating. This may be due to the large thermodynamic difference between HisMAN and PDMS, which causes the hydrophilic polymer DHM to be encapsulated within the coating and not enriched on the surface, or only a small amount is exposed on the surface (As shown in Fig. 1b, no characteristic peaks attributable to $\text{C}=\text{O}$ were observed in the coating). At the same time, the high PDMS content causes the intensity of the absorption peaks to mask the absorption peaks of DHM, resulting in the inability to observe the corresponding groups in ATR-IR.

Contact angle tests were performed on the PDMS and PDHM-x coatings to investigate the effect of DHM occupancy on the wettability

of the PDHM coatings. The surface energy of the coatings was calculated to obtain the surface energy of the coatings through the water and diiodomethane contact angles and the corresponding data are presented in Fig. 1c, d and Table S3. The PDMS coatings showed high contact angles with water and diiodomethane of 114.56° and 69.76° , respectively. When the addition of DHM was 5 wt%, the water contact angle and the diiodomethane contact angle of PDHM-5 showed a decrease to 112.03° and 67.40° , respectively. With further increase in the amount of DHM, the contact angles of PDHM-10 and PDHM-15 decreased but the surface wettability of the coatings remained hydrophobic with contact angles of 106.47° , 64.50° and 101.16° , 61.87° respectively. This indicates that due to the enrichment of high molecular weight PDMS at the air-polymer interface, the wettability of the coatings is not affected by the addition of hydrophilic polymer. In addition, the surface energy of the PDMS coating was 23.26 mN/m and those of PDHM-5, PDHM-10 and PDHM-15 were 24.44 mN/m , 26.19 mN/m and 27.86 mN/m respectively (Fig. 1d). Although the surface energy of the PDHM coating increases slowly with the addition of hydrophilic polymers (23.26 mN/m to 27.86 mN/m), it remains within the $20\text{--}30\text{ mN/m}$ range [5,36], indicating that the coating has a certain degree of fouling release capability. Therefore, adjusting the mass fraction of the DHM hydrophilic polymer enables the surface energy of the coating to be controlled to achieve fouling release while imparting other desired properties.

3.1.2. Zeta potential and surface reconstruction

As demonstrated in Fig. 1e, the ratios of HisMAN and DMAEMA were determined by zeta potential. The differing pKa values (imidazole group: 6.94, carboxylic acid group: 2.59 [37], and tertiary amine group: 7.0–7.8 [38]) result in a greater protonation capacity in HisMAN relative to DMAEMA, thus enabling its domination of the pH response process. It was determined that 50 mmol DMAEMA would be utilized as the base monomer in order to modulate the molar mass of HisMAN (0.5, 1, and 10 mmol, as Table S1) to bring the coating surface closest to electro-neutrality at pH 7.0. The prepared DHM was then applied to form amphiphilic coated PDHM with PDMS-OH at the maximum mass ratio (15 wt%), designated as PDHM_{0.5} (His/DMA = 0.5/50), PDHM₁ (His/DMA = 1/50), and PDHM₁₀ (His/DMA = 10/50), respectively [39]. The zeta potentials are illustrated in Fig. 1e, with values of 13.27 mV for PDHM_{0.5}, −15.31 mV for PDHM₁, and −32.63 mV for PDHM₁₀. This suggests that under neutral conditions, the carboxyl group undergoes strong ionisation, becoming completely deprotonated to form a

negatively charged COO[−], while the imidazole and tertiary amine groups undergo weak protonation to a lesser extent, insufficient to counteract the total negative charge of the carboxyl group. Consequently, the potential of the PDHM coating becomes more negative with increasing HisMAN occupancy. In order to ensure the electroneutrality of the coating surface, PDHM_{0.5} was tested in order to prepare PDHM coatings in a ratio to investigate the corresponding properties.

The presence of two thermodynamically different components in the amphiphilic coating results in microphase separation after forced cross-linking through chemical bonding. The outcome of this process is the formation of chemically heterogeneous surfaces at the micro-nanoscale. When alterations are induced in the surface environment of the coating, this initiates local expansion or contraction of the constituent components, and surface reconstruction ensues [40,41]. A consequent outcome of this process is a modification in the surface properties of the coating. The coatings were immersed in ASW for a period of 3 days, after which they were observed by CLSM. This was to compare the surface

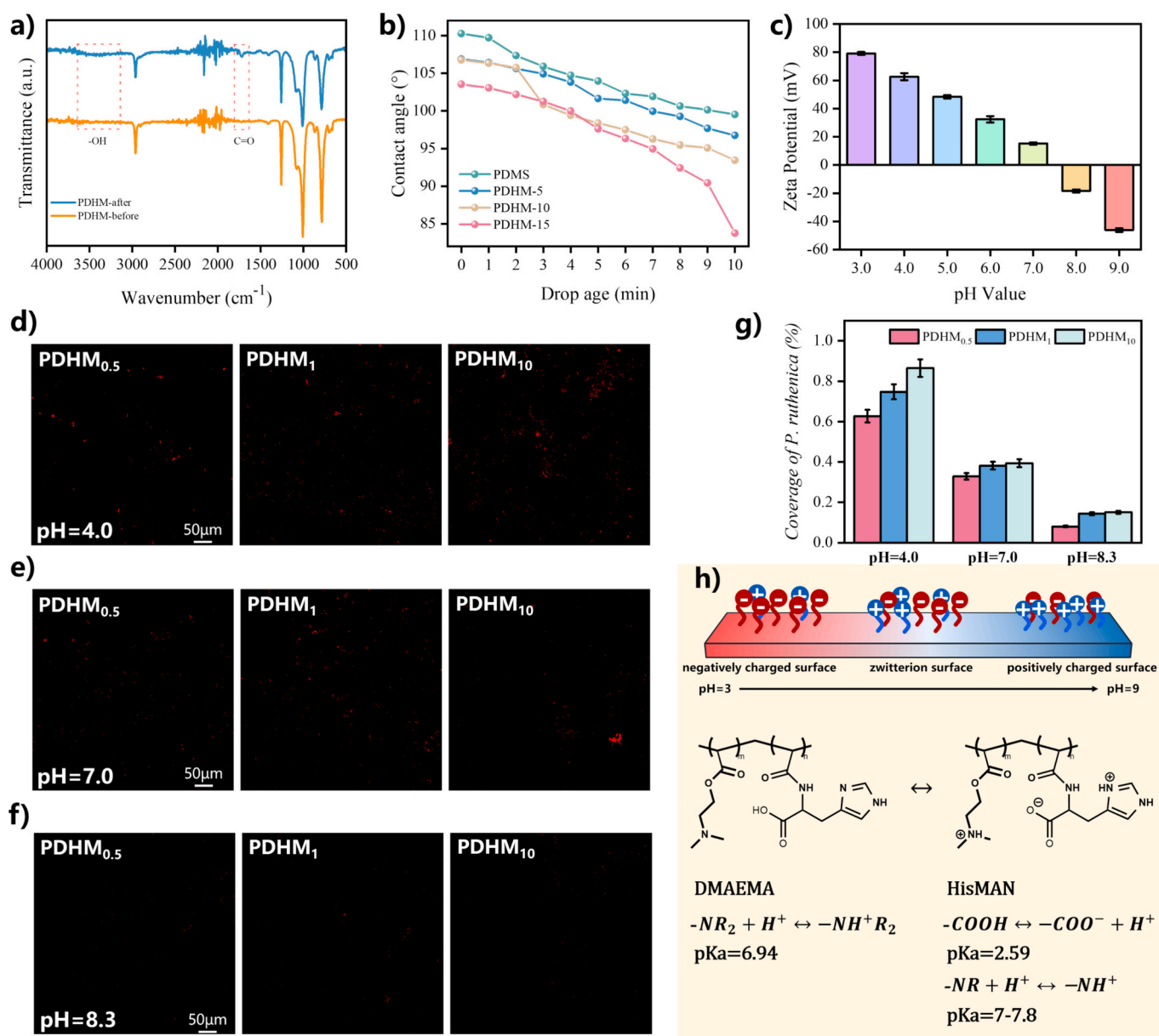


Fig. 2. (a) Comparison of ATR-IR spectra before and after immersion of PDHM-x coating, (b) Preparation of amphiphilic smart antifouling coating PDHM-x, (c) The water dynamic contact angle test of PDMS, PDHM-x coatings in 10 min. Fluorescence microscopy images of *P. ruthenica* attachment of PDHM_{0.5}, PDHM₁ and PDHM₁₀ at pH 4.0 (d), pH = 7.0 (e), pH = 8.3(f), (g) the adhesion coverage of *P. ruthenica*, and (h) schematic representation of the pH response of the PDHM.

morphology before and after the immersion of PDHM-x and PDMS, in order to explore the macroscopic changes induced by the surface reconstruction of amphiphilic coatings (Fig. 1f). As illustrated in Fig. 1f-1, a comparison is presented of the PDMS coating before and after immersion. A comparison of the 3D morphology and the physical image reveals that the morphology of the PDMS coating remains unchanged before and after the immersion process. This phenomenon can be attributed to the chemical inertness of the PDMS coating, which enables it to maintain its stability when submerged in seawater due to its inability to interact with water molecules. The incorporation of DHM (see Fig. 1f-2, 1f-3 and 1f-4) results in the emergence of disparate structural domains on the surface of the PDHM coating. The flat, smooth, light-coloured regions are composed of hydrophobic domains of PDMS, while the dark-coloured regions are hydrophilic domains of DHM. Prior to the process of immersion, the DHM manifested as a circular vesicle structure on the coating surface. As the content of DHM increases, a black, smooth pit structure gradually forms on the surface and its size increases concomitantly. This phenomenon can be attributed to the fact that hydrophilic polymers constitute a relatively minor proportion of the PDMS, and their aggregation is driven by thermodynamic incompatibility. As the solvent evaporates, the hydrophobic polymers are preferentially enriched at the air-polymer interface, while the hydrophilic polymers are encapsulated in the hydrophobic polymers, forming raised vesicles or pit-like structures. As the interface transitions to a water-polymer state, it becomes evident that an increase in dark-coloured hydrophilic domains is accompanied by a decrease in light-coloured hydrophobic domains on the surface of the coatings. Only a few black pits retain hydrophobic domains around the structure, suggesting that hydrophilic polymers migrate in response to the change in interfacial state, while hydrophobic polymers undergo shrinkage. Comparing the roughness of the coating before and after immersion (as shown in Table S4) provides a more intuitive reflection of the surface change process of the amphiphilic coating. For the PDMS coating, the Ra

value remained at 2 μm before and after immersion, while the Rq value increased slightly from 3 μm to 4 μm , which is consistent with the changes shown in Fig. 1f-1. The addition of DHM caused the Ra of the coating to increase gradually. This increased from 3 μm for PDHM-5, to 9 μm for PDHM-10, and finally to a maximum of 16 μm for PDHM-15, when the DHM content reached 15 wt%. At the same time, the Rq of the amphiphilic coating increased significantly, reaching values of 7 μm , 15 μm and 25 μm respectively. After three days of immersion in artificial seawater, the amphiphilic coatings exhibited different trends of change. The Ra and Rq values of PDHM-5 were 4 μm and 8 μm , respectively, indicating minimal change compared to the values before immersion. This is due to the lower content of the hydrophilic polymer DHM in PDHM-5 resulting in minimal impact on the coating's overall morphological changes after underwater immersion. Due to the increased DHM content, the Ra and Rq values of PDHM-10 increased significantly after immersion, reaching 12 μm and 17 μm , respectively. Interestingly, when the DHM content reaches 15 wt%, the hydrophilic polymer covers the coating surface, making it smoother. The Ra and Rq values of PDHM-15 decreased to 10 μm and 14 μm , respectively, compared to before immersion. This indicates that, although microphase separation of the hydrophilic and hydrophobic components occurs before immersion due to thermodynamic differences, increasing the roughness of the amphiphilic surface, in a seawater environment the air-coating interface is replaced by a water-coating interface. In order to maintain the minimum interfacial energy of the coating, the hydrophilic polymers rapidly migrate to the surface, thereby smoothing it. This phenomenon is a special type of surface reconstruction in amphiphilic coatings [42].

Comparing the ATR-IR patterns of the PDHM-x coating before and after immersion (Fig. 2a), the C=O stretching vibration peak at 1730 cm^{-1} and the -OH absorption peak at 3300 cm^{-1} are observed. The reason for this is that, when the interface of the coating is changed, the coating has to adjust the arrangement of polymer groups on the surface to keep the lowest interfacial energy. Therefore, when the amphiphilic

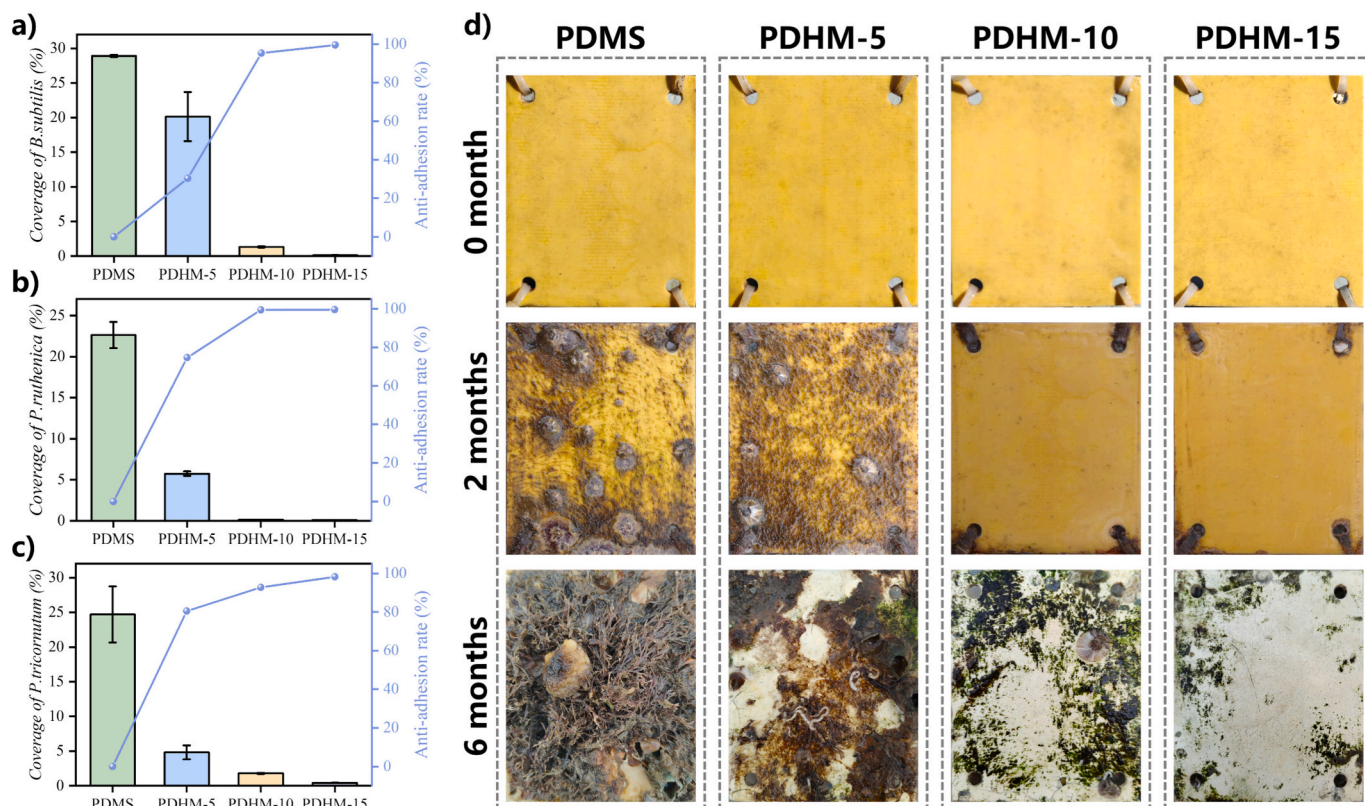


Fig. 3. (a) Preparation of pH-responsive hydrophilic polymer chain segments containing zwitterion-DMAEMA, (b) Preparation of amphiphilic smart antifouling coating PDHM-x, (c) Schematic of the anti-fouling mechanism and pH response of the PDHM coating.

coating is immersed in seawater, the hydrophilic polymer DHM migrates towards the surface and partially enriches, making the coating surface less hydrophobic to meet the lower interfacial energy. This coincides with the change in the surface morphology of the amphiphilic coating in Fig. 1f, which is further demonstration of the surface reconstruction of the PDHM coating after seawater immersion. The comparison of water contact angle changes between PDMS coating and PDHM-x coating over 10 min demonstrated the result of the migration of the hydrophilic polymer being significant in terms of changes to the surface wettability of the amphiphilic coatings (Fig. 2b). The initial water contact angles of the PDMS and PDHM-x coatings were 110.26°, 106.87°, 106.78° and 103.53°, respectively. As the water droplets spent more time on the surface, the contact angle reached 99.52° after 10 min for the PDMS coating, 96.75° for the PDHM-5 and 93.44° for the PDHM-10. The contact angles of all three decreased by approximately 10°, whilst maintaining their hydrophobic characteristics. Nevertheless, a shift in wettability from hydrophobicity to hydrophilicity was observed for PDHM-15 after a duration of 10 min, accompanied by a notable decrease in the contact angle of 19.78°, which reduced to 83.75°. This finding suggests that at 15 wt% DHM, the coating's wettability undergoes a substantial alteration following surface reconstruction, resulting in a hydrophilic surface.

The migration of hydrophilic chain segments has been demonstrated to effect alterations in both the wettability of the coating and the binding mode between the coating surface and water molecules [43]. The binding state of the water molecules on the surface was analyzed by Gaussian fitting of the ATR-IR spectra of the PDHM-x coating (see Fig. S2a-c). The results obtained demonstrated that the surface of the amphiphilic coating was composed of both free water (3400–3300 cm^{-1}) and bound water (3300–3200 cm^{-1}) [44], with the binding mode between water molecules and the coating transitioning from free water to bound water in accordance with an increase in the DHM content. Through calculation of the respective fitted areas (Fig. S2d), the percentage of bound water on the surface of the PDHM-5 coating is found to be 43.73 %, that of the PDHM-10 is 46.34 %, and that of the PDHM-15 coating is the highest attaining 57.45 %. The reason for this is that zwitterion are primarily hydrophilic monomers, meaning that they are able to form hydrogen bonds and electrostatic interactions with water molecules. Appropriate increases in content can effectively enhance the strength of bound water. Such enhancements have been demonstrated to result in improved stability of the hydrated layer, greater difficulty in the attachment of fouling organisms, enhanced fouling impedance, and improved antifouling performance of the coating [45].

3.1.3. Adhesion and impact resistance of coatings

The adhesion and impact strength of antifouling coatings directly affect their structural integrity under complex working conditions. In this regard, the improvement of adhesion and impact strength of coatings after the introduction of DHM was investigated through the carrying out of tensile testing and impact resistance testing of amphiphilic coatings [46]. As demonstrated in Fig. S3, the adhesion of the PDMS coating on the surface of the interlayer was 0.74 MPa. The adhesion of the amphiphilic coatings formed after the introduction of DHM increased in proportion to the DHM content. The adhesion of PDHM-5 was elevated to 0.93 MPa, and the adhesion of PDHM-10 and PDHM-15 reached 0.97 MPa and 1.01 MPa, respectively. Fig. S4 shows the impact resistance of the coatings. The impact strength of both PDMS and PDHM-5 was found to be 45 kg·cm. With further increases in the DHM content, the impact strength of PDHM-10 and PDHM-15 increased to 50 kg·cm and 55 kg·cm, respectively. Compared with PDMS, PDHM adhesion and impact strength are enhanced. The reason for this is that DHM contains a high number of polar groups, which can improve coating cohesion through hydrogen bonding and, at the same time, increase the strength of the bond with the substrate [47,48]. At the same time, the crosslinking density of the amphiphilic coating is increased through the condensation reaction, which ensures that the macroscopic phase

separation of hydrophilic and hydrophobic components in the coating is effectively avoided [49]. Thus, the impact strength of the coating is improved and its stable use is ensured, with premature peeling or breakage being avoided.

3.1.4. pH response of PDHM

The carboxyl groups, imidazole groups and tertiary amine groups contained in PDHM have different pKa values, which enable the coating to protonate under different pH environments and exhibit pH-responsive properties. The pH response trend of the coating was obtained by testing the surface zeta potential of PDHM at different pH values, as illustrated in Fig. 2c. The surface of PDHM exhibited a positive potential of 79.13 mV when the pH of the test solution was 3.0. As the pH increases, a decrease in the surface potential of the coating is observed, with the surface zeta potentials measuring 62.63 mV, 48.40 mV, 32.40 mV and 15.27 mV at pH 4.0, 5.0, 6.0 and 7.0, respectively. In the alkaline environment, the surface exhibited a negative zeta potential of −18.40 mV at a pH of 8.0. The lowest negative zeta potential was observed at pH = 9.0, with a specific value of −46.13 mV. In order to further investigate the effect of pH response on the antifouling performance of PDHM coatings, *P. rutenica* was chosen as the model bacteria to test the bacterial adhesion rates of PDHM_{0.5}, PDHM₁ and PDHM₁₀ at pH 4.0, 7.0 and 8.3, respectively (Fig. 2d-f). By comparing the three His/DMA PDHM coatings, it was found that at an environmental pH of 4.0 (Fig. 2d), the amount of adhesion on the surface of the PDHM_{0.5}, PDHM₁ and PDHM₁₀ coatings was low, with the lowest area of adhesion being only 0.63 % for the PDHM_{0.5} and 0.71 % and 0.86 % for the PDHM₁ and PDHM₁₀ coated surfaces, respectively. The trend was consistent when the pH was 7.0 and 8.5 (Fig. 2e and f), with PDHM_{0.5} exhibiting the lowest adhesion area of 0.33 % and 0.08 %, though this was not significantly different from the other coatings, with PDHM₁ showing 0.38 % and 0.14 % and PDHM₁₀ showing 0.39 % and 0.15 %.

It has been demonstrated that the inhibition of *P. rutenica* by PDHM is more effective in acidic environments where there is higher content of protonation forming positively charged groups, such as amine or imidazole groups (Fig. 2h). This suggests that the amount of charge on the surface of PDHM has an inhibitory effect on *P. rutenica*. This is due to the fact that a large amount of positive charge can deplete the bio-negative charge on the surface of the bacteria and prevent their normal metabolism and respiration process [50], therefore at pH = 4.0, less bacteria were adhesion to the PDHM_{0.5} surface than to the other coatings. It is noteworthy that the adhesion rate of bacteria on the surface of all three coatings decreased in accordance with rising environmental pH value. On the one hand, when the negative charge on the surface was enriched by pH elevation, the antifouling mechanism of the coatings against bacteria was converted from the effect of charge to the impedance effect through electrostatic repulsion and hydration layer, and this synergistic effect could effectively reduce the colonization of *P. rutenica* on the surface. On that hand, when the coating surface is enriched with positive charges, bacteria can improve their adhesion on the surface through electrostatic action [51,52]. Therefore, it can be found that the bacterial attachment area decreases and tends to be the same for all three coatings with increasing pH.

In summary, the PDHM coating prepared with HisMAN/DMAEMA = 0.5 can achieve efficient antifouling in complex marine environments through the synergistic effect of pH response and fouling release-fouling impedance of the amphiphilic coating.

3.2. Antifouling properties of coating

Marine fouling is a complex process involving the adhesion of microorganisms, microalgae and crustaceans to surfaces. When these organisms colonise the surface of a membrane made of organic matter, they release EPS to form a biofilm [53]. This further promotes the adhesion of various macroscopic marine fouling organisms, such as barnacles and mussels, to the surface of the biofilm. Forming a sediment

layer that is difficult to remove, the result is marine fouling. Therefore, *P. ruthenica* and *B. subtilis* were chosen as model bacteria and *P. tricornutum* as model diatoms, respectively, and combined with marine fouling experiments to verify the anti-fouling performance of the PDHM coating.

3.2.1. Anti-bacterial adhesion properties

Fig. S5 shows fluorescence microscopy images of *B. subtilis* adhesion to PDMS and PDHM-x coatings. The fluorescence intensity of the four sample surfaces is significantly distinct, with large numbers of bacteria adhering to PDMS and PDHM-5 surfaces and weaker fluorescence on PDHM-10 and PDHM-15 surfaces. This suggests that PDHM-10 and PDHM-15 coatings have an inhibitory effect on *B. subtilis* adhesion. The specific adhesion coverage and anti-adhesion rate are shown in Fig. 3a, which shows that the adhesion coverage of the PDMS coating is 28.90 %. The adhesion coverage of PDHM-x gradually decreased with the increase of DHM content, which was 20.14 % (PDHM-5), 1.34 % (PDHM-10) and 0.13 % (PDHM-15), respectively. Correspondingly, the anti-adhesion rate of PDHM-5 was 30.31 %, PDHM-10 was 95.37 %, and PDHM-15 had the highest anti-adhesion rate of 99.54 % (Table S5).

The adhesion of *P. ruthenica* to the four coated surfaces was similar to that of *B. subtilis* (see Fig. S5). The highest adhesion coverage was 22.62 % on the PDMS surface and 5.74 % on the PDHM-5 surface. When the DHM percentage was greater than 10 wt%, the adhesion coverage decreased significantly, reaching 0.13 % for PDHM-10 and 0.09 % for PDHM-15. Fig. 3b shows the anti-adhesion rate of the PDHM-x coating on *P. ruthenica* as 74.63 % for PDHM-5, 99.41 % for PDHM-10 and 99.59 % for PDHM-15 (Table S5). When comparing the adhesion of *P. ruthenica* and *B. subtilis* on the PDHM surface, it was observed that, when the DHM percentage is greater than 10 wt%, the PDHM-x coating effectively reduces bacterial adhesion on the coated surface, with PDHM-15 exhibiting the highest anti-adhesion rate.

3.2.2. Anti-algae adhesion properties

The anti-algal adhesion performance of PDMS and PDHM-x was evaluated by comparing the diatom adhesion on their surfaces (Fig. S5 and 3c). Through the comparative analysis of the adhesion properties of *P. tricornutum* on surfaces treated with PDMS and PDHM-x coatings, it was determined that the adhesion coverage of PDHM-x coating was significantly lower than that of PDMS coating. The adhesion coverage observed on the surfaces of the four groups of samples were found to be 24.69 %, 4.81 %, 1.80 %, and 0.44 %, respectively. Furthermore, the anti-adhesion rates of PDHM-5, PDHM-10 and PDHM-15 coatings were 80.51 %, 92.73 % and 98.23 %, respectively (Table S5). The cell wall of *P. tricornutum* contains a significant quantity of silica, which facilitates its adhesion to the surface of silicone-based FRCs [54]. The hydrophilic polymer DHM introduces a chemical heterogeneous structure to the coating surface, thereby enhancing its anti-diatom adhesion performance. Consequently, the surface adhesion of PDHM-15 to *P. tricornutum* is significantly lower than that of PDMS.

3.2.3. Marine field antifouling properties

The complexity and variability of the marine environment have a significant impact on the antifouling performance of the coatings. In order to verify the antifouling performance of the PDHM-x coatings and the correspondence law, the PDMS coatings and the PDHM-x coatings were placed on a floating raft in the sea of Xiamen, China, at Daliapuyu Island (24°33'N, 118°09'E) to conduct an antifouling performance test for 180 days in the real sea, as shown in Fig. 3d.

Subsequent immersion of the coatings for the 2 months period resulted in a certain quantity of barnacles adhering to the surface of the PDMS and PDHM-5 coatings. The surface of the PDMS coatings demonstrated the highest rates of adherence, and a substantial amount of marine sludge was deposited. However, the surface of the PDHM-10 and PDHM-15 coatings exhibited no signs of fouling organisms. Following the 6 months submersion, a substantial quantity of algae,

duster, and crustaceans, including barnacles, was observed on the surface of the PDMS coating. This resulted in an antifouling performance score of 0, indicating a complete failure of the antifouling coating. The PDHM-5 coating exhibited a limited adherence of barnacles and tube-worms, accompanied by cracking along its edges, resulting in an antifouling performance rating of 48. The PDHM-10 coating exhibited substantial coverage of algal colonization on its surface, accompanied by a limited attachment of barnacles, which resulted in score of 78. In contrast, the PDHM-15 coating exhibited a negligible presence of algae and no evidence of crustaceans such as barnacles, thereby achieving an antifouling performance score of 94.

The antifouling performance of the PDHM-15 coating has been demonstrated to be effective in a marine environment through comparative analysis.

3.2.4. Impact of PDHM on the early microbial succession dynamics

PDHM utilizes a combination of surface charge conversion via pH response and the inherent fouling impedance-fouling release of amphiphilic coatings to provide PDHM-15 with excellent fouling resistance. However, the differences in the effects of this synergistic mechanism on fouling organisms compared to PDMS have yet to be adequately explored. In this regard, the adherent bacteria on the coated surface were selected as the subject of the research, and the changes in the bacterial community on the two different coated surfaces were analyzed using 16S rDNA technology to determine the difference in antifouling behaviour between the amphiphilic coatings and PDMS. To avoid convergence of bacterial communities on coated surfaces, the test period was set at 14 days [55]. Samples were taken on the first, third, seventh and fourteenth days to analyze the surfaces. Surface samples obtained within the PDMS on days one, three, seven and fourteen were labelled S-1D, S-3D, S-7D and S-14D, respectively. PDHM surface samples were labelled H-1D, H-3D, H-7D and H-14D.

Bacterial dynamics were analyzed using operational taxonomic units (OTUs). All valid sequences were clustered into 332–1718 OTUs at 97 % similarity (see Table S6). The rarefaction curves and rank-abundance curves of all samples showed a plateau ranging (Fig. S6a-d), indicating that the obtained sequences adequately represent the diversity of the microbial community in each plate group. Fig. 4a-1 and b-1 show the sobs curves and shannon curves for the α diversity of PDMS and PDHM. Over a 14-day period, it can be seen that both coating diversity curves follow the same trend, with the correlation values detected on the surface decreasing from day 3 onwards, reaching a minimum on day 7, before showing an increasing trend on day 14. In addition, comparing the variability of PDMS and PDHM at the four sampling times (see Fig. 4a-2 and b-2), it can be seen that there was less differentiation in the first three sampling times ($p > 0.5$). This suggests that the number of bacteria detected adhering to the surfaces of the PDMS and PDHM were similar in terms of number, richness and evenness before the 7-day period. By day 14, there was still no significant difference in the shannon index between the two coatings, and bacterial richness and evenness were similar. However, the p -value of the sobs index was 0.0113, much smaller than 0.5 and indicating a significant difference in the number of OTUs detected on the PDHM and PDMS surfaces. Among them, the PDMS surface OTUs reached 869 and the PDHM surface OTUs were 633. As the bacterial adhesion time to the coating surfaces increased, it was obvious that the degree of bacterial dispersion on the two surfaces remained similar. However, the number of OTUs on the surface of PDMS was higher than that on the 14th day of PDHM, which indicated that the surface of PDMS had more species and was evenly distributed, while the surface of PDHM had fewer species and was evenly distributed, and both surfaces maintained stable diversity. Furthermore, this suggests that the two coatings have similar antifouling mechanisms and that PDHM is more resistant to bacterial adhesion than PDMS.

The difference in diversity between the two coatings was primarily influenced by their surface properties. The species composition of PDMS and PDHM surfaces was analyzed to investigate whether surface

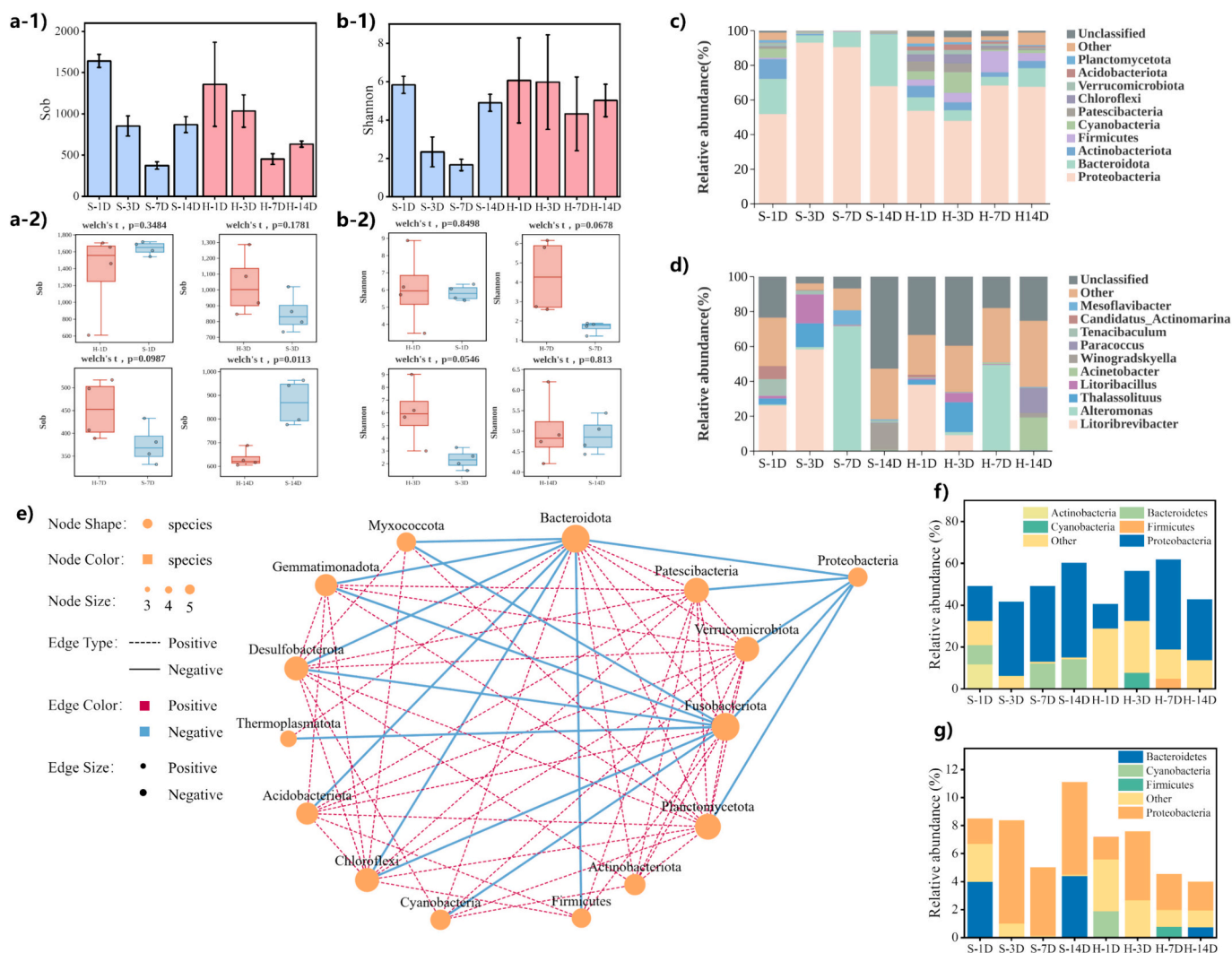


Fig. 4. (a-1 and b-1) α diversity index and (a-2 and b-2) α diversity statistical test for PDHM and PDMS, (c) Phylum-level and (d) genus-level classification of surface bacteria, (e) Species correlation network diagram between bacteria on coated surfaces, and contribution of stacked maps to (f) film formation and (g) MGEs.

properties affect bacterial species. At the phylum level (Fig. 4c and Table S7), the taxonomic composition of bacteria attached to the surface of PDMS was assessed. It was found that the Proteobacteria was dominant in all four groups (S-1D, 51.83 %; S-3D, 93.04 %; S-7D, 90.47 %; and S-14D, 67.95 %), especially in S-3D and S-7D, where it exceeded 90 %. In S-1D and S-14D, its share had decreased, but remained above 50 %. With the exception of Proteobacteria, the Bacteroidetes was the second most prevalent in the four groups, attaining 20.32 %, 4.22 %, 8.76 %, and 29.86 %, respectively. Moreover, the proportion of Actinobacteriota and Firmicutes in S-1D was significantly higher than in the other three groups (11.39 %, 0.79 %). It can be found that PDMS has a propensity to adhere to surfaces in the presence of one or more dominant bacterial strains, with this propensity increasing with increasing immersion time. The overall trend of bacterial attachment on the PDHM and PDMS surfaces was similar, but there were some differences. Despite the higher occupancy of Firmicutes and Cyanobacteria, other bacterial taxa did not become extinct as time progressed. On days 3 and 7, the proportions of Firmicutes and Cyanobacteria exceeded those of Bacteroidota (H-3D, 11.89 % for Cyanobacteria and 6.12 % for Bacteroidota; H-7D, 12.21 % for Firmicutes and 5.02 % for Bacteroidota). By conducting a thorough analysis of the bacteria present on the coated surface at the genus level, it was ascertained that the top 10 in terms of adherent bacterial richness were *Litoribrevibacter*, *Alteromonas*, *Thalassolituus*, *Litoribacillus*, *Acinetobacter*,

Paracoccus, *Winogradskyella*, *Tenacibaculum*, *Mesoflavibacter* and *Candidatus Actinomarina* (Fig. 4d). Surprisingly, the vast majority of the dominant genera on the PDMS and PDHM surfaces, with the exception of the S-14D surface where the dominant bacterium was *Winogradskyella* (Bacteroidota, Bacteroidia), were attributed to Proteobacteria, Gammaproteobacteria (Table S8). The attachment of bacteria to PDMS and PDHM surfaces is consistent with existing reports [56,57] that Proteobacteria, Bacteroidota, Actinobacteriota and Firmicutes are the bacteria with the highest attachment rates to structure surfaces. Among them, many species of Proteobacteria have motile structures such as flagella, which can attach to most material surfaces [58,59], through group sensing, increase colonization efficiency [60]. Therefore, although PDHM inhibits bacteria more than PDMS, Proteobacteria still adhere to its surface.

Furthermore, as shown in the species correlation network diagram (Fig. 4e), Proteobacteria showed lower connectivity with other bacteria, with negative correlations only with Bacteroidota, Patescibacteria, Verrucomicrobiota, Fusobacteriota, and Planctomycetota. This means that there is only a competitive relationship between Proteobacteria and other bacteria. In contrast, the majority of bacteria show a positive correlation with each other, and there are also bacteria that have both synergistic and competitive relationships, such as Bacteroidota. The correlation between the bacteria revealed that Proteobacteria, as the dominant bacteria at the initial stage of attachment, compressed the

growth resources of other bacteria through the competitive relationship with other bacteria over time, resulting in the tendency of bacteria to be homogenized, with the highest percentage of attachment on the surface of PDMS and PDHM [61]. The EPS (e.g. polysaccharides, proteins, DNA, etc.) released by the bacteria after biofilm formation would increase the complexity of the community structure and improve the adaptation and stability of the bacteria to the environment [62]. Among these, mobile genetic elements (MGE), a class of DNA sequences that can be transferred within the genome or between different organisms, reflect the environmental adaptation of bacterial communities and interspecies interactions [63,64]. Therefore, when comparing the film-forming properties and the relative abundance of mobile genetic elements of bacteria on different surfaces, the strength of bacterial inhibition by the coating can be visualized. Functional analyses of membrane-forming and MGEs of bacteria on the surface of PDMS and PDHM revealed that the membrane-forming bacteria on the surface of PDMS were mainly Proteobacteria and Bacteroidota, and the membrane-forming bacteria on the surface of PDHM were Proteobacteria (Fig. 4f), and that the relative abundance of membrane-forming bacteria on the surface of PDMS was higher than that of PDHM over time. Furthermore, the relative abundance of bacteria releasing MGEs was higher on PDMS compared to PDHM (Fig. 4g). In the same marine environment, a more pronounced difference in the bacterial community only occurred when the substrate to which the bacteria were attached had some effect on the bacteria. Therefore, the PDHM surface inhibits the film-forming properties of bacteria and reduces the release of MGEs, resulting in reduced environmental adaptation of the bacterial community surface, which is attributed to the specific antifouling mechanism of PDHM.

The PDHM-x coating operates as an intelligent and highly efficient antifouling system, leveraging the synergistic interplay between hydrophilic and hydrophobic components alongside a pH-responsive surface charge switching mechanism. In the marine environment, under conditions free of fouling organisms, the coating surface is exposed to a weakly alkaline milieu, prompting deprotonation of the carboxylic acid (-COOH) groups and yielding a negatively charged interface. This electronegative surface, in concert with a robust hydration barrier, generates synergistic antifouling effects: electrostatic repulsion and hydrated layer resistance collectively elevate the energetic barrier to biofouling adhesion, thereby effectively inhibiting colonization. Should fouling organisms manage to breach these initial defenses and adhere to the surface, their metabolic excretion of acidic substances locally alters the interfacial pH. This triggers a reversible protonation response in both amine (-NH₂) and carboxylic acid functional groups, converting the surface charge from negative to positive. The resulting positively charged interface is capable of suppressing biological activity and inducing cytotoxic effects, thereby countering further fouling progression. Simultaneously, the inherent low surface energy contributed by hydrophobic segments facilitates fouling release, enabling effortless removal of adhered organisms under hydrodynamic flow conditions. Once the coating is re-immersed in seawater, the surface pH swiftly resets to ambient alkaline levels, restoring the deprotonated state of the carboxylic groups and regenerating the negatively charged antifouling surface. This dynamic transition re-establishes the electrostatic-hydration dual barrier, enabling continuous self-adaptation to fouling threats. Consequently, the PDHM-x coating achieves a cyclic “repel-resist/destroy-release” antifouling sequence, ensuring persistent and adaptive protection against biofouling across varying marine conditions.

4. Conclusion

This study establishes pH-responsive amphiphilic coatings as a potent marine antifouling strategy. We engineered the PDHM coating by integrating zwitterionic-tertiary amine polymer chains (DHM) exhibiting pH-responsive hydrophilicity within a PDMS matrix. This design enables intelligent, environment-responsive surface charge modulation via protonation/deprotonation, conferring exceptional and long-lasting

antifouling performance. At an optimal DHM content of 15 wt%, PDHM-15 achieved outstanding anti-adhesion rates of 99.54 % against *B. subtilis* and 99.59 % against *P. rutenica*, alongside a 98.23 % reduction in *P. rutenica* adhesion. Remarkably, PDHM-15 maintained high efficacy throughout a rigorous 6-month marine field trial. Mechanistic insights from 16S rDNA analysis revealed PDHM's capacity to suppress bacterial biofilm formation, reduce the release of MGEs, and significantly decrease surface bacterial adhesion rates. These effects arise from the coating's unique “repel-repel/destroy-remove” cycle, which underpins its superior antifouling capability. Integration of high-throughput sequencing elucidated these fundamental mechanisms, paving the way for intelligent, next-generation fouling-control systems.

CRedit authorship contribution statement

Chuaning Wei: Writing – review & editing, Writing – original draft, Conceptualization. **Xianhui Zhang:** Writing – original draft, Investigation. **Yajun Deng:** Writing – original draft, Investigation. **Bo Wu:** Writing – review & editing, Investigation. **Jianhua Wu:** Writing – review & editing, Supervision, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.porgcoat.2025.109701>.

Data availability

The authors are unable or have chosen not to specify which data has been used.

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